

(TR-104) PYTHON WITH AI/ML

Training Day 39 Report

Date: 5 March 2026

Topic Covered: Secure RAG Pipeline Development with Guardrails & Safe Retrieval

The thirty-ninth day of training focused on extending the RAG-based AI Assistant project by implementing secure retrieval mechanisms, integrating guardrails, and building a safe context-aware response generation pipeline using Ollama and the Mistral model.

- Used embedding manager to generate query embeddings for semantic vector search.
- Connected to ChromaDB vector database for retrieving top-k relevant document chunks along with metadata.
- Created a reusable retriever class for efficient document retrieval workflows.
- Added system prompts to ensure the model answers strictly from the provided document context.
- Implemented deterministic guardrails to block unsafe or restricted queries such as hacking and exploit-related requests.
- Added model-based guardrails to detect sensitive outputs including PII and confidential information.
- Used the Mistral model locally through Ollama for response generation.
- Built a complete RAG pipeline integrating retrieval, guardrails, and safe answer generation.
- Developed a reusable RAG chain function for secure querying and response handling.
- Tested the system with sample queries to verify safe, context-aware, and retrieval-based responses.
- Strengthened understanding of secure AI system design and guardrail implementation in production-style RAG systems.
- Learned how retrieval systems, local LLMs, and safety mechanisms work together in modern AI assistant architectures.

- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/projects/blob/main/src/retriever.py>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/guardrails-in-ai-guiding-intelligence-safely-6baa2c7e6da9>